

Testing Surprisal Theory with Language Models

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Project Repository: [GitHub Link](#)

1 Goal

This project aims to investigate how computational models of language processing can explain human sentence comprehension, with a particular focus on the relationship between linguistic surprisal and reading time. The study evaluates whether probabilistic expectations generated by language models correspond to the cognitive effort humans experience while reading. By comparing lexical (trigram), syntactic (PCFG), and neural (GPT-2) language models I seek to examine how different forms of contextual prediction relate to observable behavioral measures such as reading time. The work combines computational modeling, statistical analysis, and psycholinguistic interpretation to bridge insights from natural language processing and cognitive science.

2 Theoretical Background

Human language comprehension involves continuously predicting upcoming linguistic input during reading and sentence processing. A central question is whether computational language models can capture the same predictive mechanisms that humans use during comprehension. Behavioral measures such as reading time and eye movements provide observable evidence of cognitive processing difficulty and are therefore widely used to evaluate theories of language comprehension.

One of the most influential frameworks in psycholinguistics is *Surprisal Theory* proposed by Levy (2008). According to this theory, the processing difficulty of a word depends on how unexpected it is in a given context. Surprisal is defined as:

$$\text{Processing Difficulty} \propto -\log P(w \mid \text{context})$$

Words with lower probability have higher surprisal and are expected to produce greater processing difficulty and longer reading times. Demberg and Keller (2008) showed that eye tracking and reading time measures reflect this cognitive complexity and can therefore be used to evaluate computational models of comprehension.

Later work by Futrell, Gibson, and Levy (2020) extended surprisal based approaches by incorporating memory constraints into sentence processing. Their work emphasized that comprehension depends not only on predictability but also on the ability to retain contextual information across long range dependencies.

Motivated by these theoretical developments, the project investigates whether language model surprisal correlates with human reading time during sentence comprehension. The project compares lexical, syntactic, and neural language models to determine which model best predicts human processing behavior. In addition, the study explores discussions about Information Locality Hint and the corresponding patterns of cognitive difficulty during reading.

3 Dataset

The project uses the *Natural Stories Corpus*, a widely used dataset for studying human sentence comprehension. The corpus contains English narrative texts designed with linguistically complex constructions such as rare syntactic patterns, ambiguities, and long range dependencies, making it suitable for evaluating processing difficulty during reading.

The dataset provides word level behavioral measures, including human reading times collected through self paced reading and eye tracking experiments which serve as indicators of cognitive effort. This combination of naturalistic text and detailed reading time annotations makes it appropriate for evaluating whether surprisal estimates from language models correlate with human processing difficulty.

4 Methodology

The objective of this methodology is to provide a rigorous framework for testing *Surprisal Theory*, which quantifies cognitive processing difficulty as a function of probabilistic unpredictability. By aligning computational estimates with human behavioral data from the *Natural Stories Corpus*, this pipeline evaluates how effectively different linguistic abstractions (lexical, structural, and neural) mirror the human reading process. The framework further incorporates the *Information Locality Hint* (ILH) to determine if word position and integration distance act as secondary modulators of processing effort.

The implementation follows a four-stage pipeline: modeling, calculation, data alignment, and statistical inference.

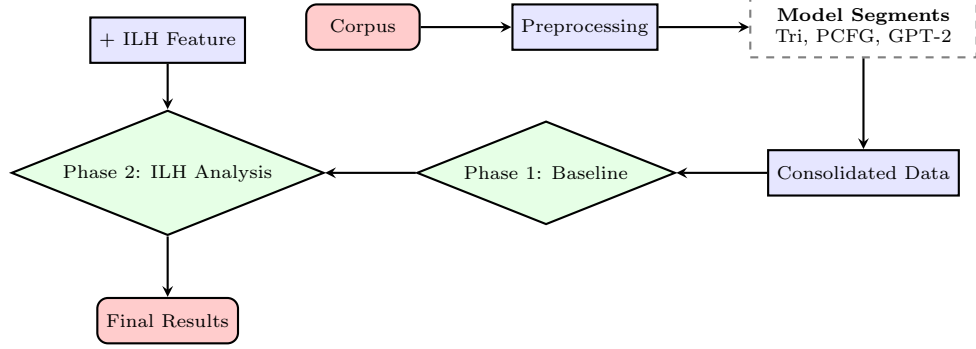


Figure 1: Computation Pipeline: Sequential flow from data ingestion to Baseline analysis, wrapping into the ILH-extended Bayesian framework.

- **Surprisal Computation:** For every word w in the corpus, surprisal is calculated as the negative log-probability given its preceding context:

$$Surprisal(w) = -\log P(w \mid \text{context})$$

- **Modeling Segments:**

- **Lexical (Trigram):** Uses local word-to-word dependencies where the probability is estimated via Laplace (add-one) smoothing to handle zero-frequency events:

$$P(w_i \mid w_{i-1}, w_{i-2}) = \frac{\text{count}(w_{i-2}, w_{i-1}, w_i) + 1}{\text{count}(w_{i-2}, w_{i-1}) + V}$$

where V represents the vocabulary size.

- **Structural (PCFG):** Estimates syntactic surprisal by inducing a grammar from the Penn Treebank and calculating the probability of the most likely parse (Viterbi) for the current sentence prefix.
- **Neural (GPT-2):** Leverages transformer-based self-attention to generate contextual representations. For words split into multiple sub-tokens $(t_1, t_2, \dots, t_n)$, the chain rule is applied to maintain the probability of the original word:

$$Surprisal(w) = \sum_{j=1}^n -\log P(t_j \mid \text{context}, t_{1 \dots j-1})$$

- **Information Locality Hint (ILH):** To account for memory decay and integration effort, a `locality_hint` was tracked, representing the distance (cumulative word count) from the start of the story item:

$$\text{Locality}(w) = \text{cumcount}(w) \in \text{Item}_j$$

- **Data Alignment:** Behavioral reading times were cleaned and log-transformed ($\log RT$) to stabilize variance. For the Bayesian models, all features were standardized for comparative stability:

$$Z = \frac{x - \mu}{\sigma}$$

This methodology concludes with a dual-evaluation strategy to validate model performance. First, **Pearson** (r) and **Spearman** correlations are computed to identify linear and rank-order relationships between surprisal and reading time. Second, a **Bayesian Linear Regression** using the No-U-Turn Sampler (NUTS) estimates the specific contribution of surprisal and locality to human processing speed:

$$\log RT_{\text{std}} \sim \text{Normal}(\alpha + \beta_{\text{surp}} \cdot \text{Surprisal}_{\text{std}} + \beta_{\text{loc}} \cdot \text{Locality}_{\text{std}}, \sigma)$$

By analyzing the posterior distributions (β coefficients) across the Lexical, Structural, and Neural segments, this research provides a definitive measure of how different forms of linguistic context influence the human cognitive bottleneck.

5 Results and Key Findings

5.1 Comparative Surprisal Analysis

- The study evaluated the predictive accuracy of word-level surprisal across lexical, structural, and neural modeling paradigms to determine which architecture best aligns with human cognitive effort.

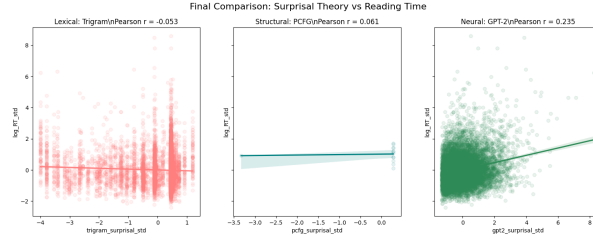


Figure 2: Final Comparative Surprisal Analysis

- **Neural (GPT-2, Green):** This model demonstrated the strongest alignment with human behavior, yielding a **Pearson correlation of 0.2479** and a Spearman rank correlation of 0.2015.
- **Structural (PCFG, Teal):** The syntactic model showed a marginal positive relationship, with a Pearson r of 0.0611 and a Spearman correlation of 0.0772.

- **Lexical (Trigram, Red):** The trigram model produced a slight negative Pearson correlation of -0.0535, indicating a limited capacity to capture cognitive load compared to higher-order models.
- The results confirm the hypothesis that **neural language models**, which integrate long range context, are significantly more cognitively plausible than simple lexical or purely syntactic models for predicting reading behavior.

5.2 Bayesian Regression and Locality Effects

- Beyond raw predictability, Bayesian inference was used to quantify the specific influence of surprisal (β_{surp}) and word locality (β_{loc}) on processing difficulty.

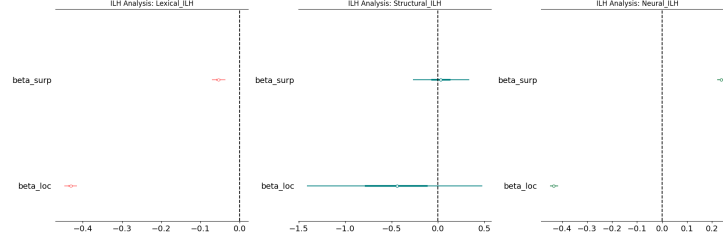


Figure 3: Effect of ILH on each model

- **Neural ILH (GPT-2):** Even when accounting for locality, GPT-2 maintains a strong positive β_{surp} of **0.248**. This proves that neural surprisal captures unique cognitive information not merely explained by word position.
- **Lexical ILH (Trigram):** The β_{surp} for this segment is near zero (-0.053), confirming that lexical surprisal alone is an insufficient predictor.
- **Structural ILH (PCFG):** PCFG surprisal and locality show high correlation ($\beta_{\text{surp}} = 0.056$), as complex structures naturally accumulate as a sentence progresses.
- **Locality Effect (β_{loc}):** The ILH revealed a significant negative effect ($\beta = -0.433$), indicating that as words progress within a story, reading times consistently decrease due to contextual accumulation.
- Human processing is governed by dual constraints: word unpredictability and the memory cost associated with discourse position.

Parameter	Mean	SD	HDI 2.5%	HDI 97.5%	R-hat
β_{surp}	0.238	0.009	0.221	0.255	1.0
β_{loc}	-0.433	0.009	-0.449	-0.416	1.0
Intercept	0.000	0.009	-0.017	0.016	1.0
σ	0.870	0.006	0.858	0.882	1.0

Table 1: Combined Bayesian Model Parameters (GPT-2 + ILH)

Model	Pearson r	Spearman ρ
Trigram	-0.0535	0.0029
PCFG	0.0611	0.0772
GPT-2	0.2479	0.2015

Table 2: Comparative Model Correlations

5.3 Statistical Summary Tables

5.4 Error Analysis and Model Limitations

- Error analysis was performed to isolate words with the highest residuals between predicted and actual reading times to identify where models fail to mirror human cognition.
 - **High-Error Constructions:** Complex tokens such as "years.", "tulip", and numerical compounds like "thirty-seven" were identified as the most difficult for GPT-2 to predict accurately.
 - **Residual Magnitudes:** Top difficult words showed error values ranging from **5.46 to 8.47**, highlighting a disconnect for specific lexical categories or punctuation.
- While neural models excel at general context, they struggle with low-frequency nouns and numerical syntax, suggesting a need for specialized modeling to capture these cognitive loads.

6 Interpretability of Results

The interpretability of the results becomes stronger when moving from simple correlation analysis to a **multivariate Bayesian framework**. While correlations provide a general relationship between surprisal and reading time, Bayesian regression allows the independent effects of **predictability** and **memory-related locality constraints** to be separated more clearly.

Analysis of the posterior coefficient distributions (β values) shows that **neural language models are not merely capturing positional effects**. Even after controlling for the Information Locality Hint (ILH), the GPT-2 model

Table 3: Top 10 Difficult Words for Prediction [cite: 287-292]

Word	GPT-2 Surprisal	$\log RT$	Error
years.	5.660709	6.772066	8.471779
tulip	7.089891	6.684828	7.573270
thirty-seven,	16.132424	6.759384	7.524738
bones?’	10.829632	6.685050	7.277805
hedge-king!’	2.975376	6.520055	6.418195
claims,	2.445083	6.464134	5.957295
noblewoman.	12.577720	6.543612	5.866331
forty-seven.	9.615927	6.480110	5.530637
air	4.980031	6.438973	5.529307
eighty-six-year-old	16.468129	6.532867	5.460214

maintains a strong positive surprisal coefficient (β_{surp}), indicating that its contextual representations encode **cognitively meaningful expectations** during reading.

The variation in β values across models also reveals a **hierarchy of cognitive plausibility**. In the lexical trigram model, the surprisal effect weakens considerably after locality information is introduced, suggesting that local lexical probability alone does not adequately explain reading difficulty. In contrast, the neural GPT-2 model shows that **both surprisal and locality independently contribute** to processing effort. This indicates that human comprehension depends on both **predictive expectation** and **memory-based integration costs**.

Residual analysis further highlights specific limitations of the models. High residuals were observed for numerical expressions such as “thirty-seven” and low-frequency nouns such as “tulip,” suggesting that neural models may still struggle with certain lexical categories and localized integration effects despite capturing broader contextual structure effectively.

Overall, the results suggest that cognitively plausible language modeling requires combining **surprisal-based prediction** with **locality and memory constraints**. Models that incorporate both factors provide a more interpretable explanation of human reading behavior and sentence comprehension.

7 Code Analysis

7.1 Dataset Loading and Preprocessing

- Before computing surprisal, the raw behavioral and linguistic data must be aligned to ensure that every word corresponds to a valid human reading time measure.
 - **Data Ingestion:** Loads the *Natural Stories Corpus* via `processed.RTs.tsv` and `processed.wordinfo.tsv`.

- **Cleaning:** Normalizes text (lowercase, punctuation removal) and filters reading times (RT) to the standard 100ms–2000ms range.
- **Consolidation:** Uses a **Left Join** anchored to the GPT-2 dataset to preserve the maximum token count (10k) and fix the "inner join bottleneck" identified in earlier pipeline iterations.
- This stage results in a standardized dataset with $\log RT$ and Z -scored features that allows for statistically stable comparisons across all modeling segments.

7.2 Model Pipelines and Surprisal Extraction

- The project requires extracting probability estimates from three distinct levels of linguistic representation: lexical, structural, and neural.
 - **Trigram (Lexical):** Uses a `defaultdict` to count (w_{i-2}, w_{i-1}, w_i) occurrences and applies **Laplace Smoothing** to compute $P(w_i | w_{i-1}, w_{i-2})$.
 - **PCFG (Structural):** Employs a `ViterbiParser` on a grammar induced from the Penn Treebank to calculate prefix probabilities.
 - **GPT-2 (Neural):** Leverages `GPT2LMHeadModel` and the chain rule to handle sub-word tokenization: $Surprisal(w) = \sum -\log P(t_j | \text{context})$.
- By automating surprisal extraction across these architectures, the code provides the necessary independent variables to test the Surprisal Theory hypothesis.

7.3 Statistical Evaluation (Phase 1: Combined Baseline)

- The initial evaluation objective is to determine which model segment most closely aligns with the linear trends of human reading behavior.
 - **Baseline Comparison:** Generates regression plots and calculates **Pearson r** and **Spearman** correlations for the Lexical, Structural, and Neural segments.
 - **Residual Analysis:** The `identify_difficult_sentences` function calculates errors between predicted and actual RTs to find high-difficulty linguistic constructions.
- This phase establishes a baseline for cognitive plausibility, identifying which modeling paradigms capture the most significant variance in reading times.

7.4 Statistical Evaluation (Phase 2: Extended ILH Analysis)

- To refine the baseline, the analysis must account for memory and integration effects as suggested by the *Information Locality Hint* (ILH).
 - **Feature Integration:** Integrates the `locality_hint` feature, representing cumulative word count per story item.
 - **Bayesian Inference:** Executes a multivariate Bayesian regression using **NUTS sampling** to estimate β_{surp} and β_{loc} simultaneously.
- This final stage quantifies the degree to which human processing is influenced by predictability versus integration distance, providing a complete information-theoretic profile of the comprehension process.

8 Conclusion

This project evaluated whether computational language models can explain human sentence comprehension through *Surprisal Theory*. By comparing Trigram, PCFG, and GPT-2 models on the *Natural Stories Corpus*, the study showed that neural language models align most strongly with human reading behavior, supporting the role of contextual prediction in comprehension.

The results also demonstrate that reading difficulty is influenced not only by surprisal but also by locality and memory-related integration effects. Bayesian analysis showed that GPT-2 surprisal remains a significant predictor even after controlling for locality, suggesting that neural language models capture cognitively meaningful expectations during reading.

Overall, the project highlights the importance of combining prediction and memory constraints to better model human language processing and demonstrates how computational linguistics and cognitive science can be integrated through behavioral analysis.